

Choosing Evidence for a Small Frozen Video-QA Model

A two-path, fully local framework for long-video question answering: design, dataset construction, learned and rule-based frame selection, and what the experiments do and do not show

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Abstract

We build a question-answering system for long videos that runs on one 16 GB consumer GPU. Instead of showing a video to a vision-language model, it retrieves the relevant transcript windows, proposes candidate frames along two paths (transcript "hot moments" and a sparse visual scan), selects four frames, and asks a frozen Qwen3-VL-2B model to answer from those frames and a short transcript excerpt. Version 1 scores **41.6%** on 440 LongVideoBench questions with MobileCLIP frame selection, +9.1 points over transcript-only. Version 2 asks whether small learned selectors, trained on measured answer changes of the frozen model, choose better frames than simple rules. On a lecture dataset we built from 20 MIT OpenCourseWare lectures, they do not: four families of learned or feature-based selectors, including one trained on deployment-matched labels, all fail to beat zero-shot transcript relevance, although an exhaustive oracle shows **+6.6 points** of headroom in the last frame slot. A visual-scan change that helped on lectures did not transfer to LongVideoBench. We also found and corrected a dataset flaw: answerer-conditioned filtering amplified option-position bias, so that "always answer B" would have outsourced every system; circular evaluation raises all lecture accuracies by about 15 points while preserving selector rankings. We report these results, the corrected numbers, and the remaining open questions without claiming a new method.

1. What this report claims

Claim	Evidence	Strength
A working, fully local long-video QA framework with per-stage cost accounting	v1 on LongVideoBench, 440 questions; $\approx$ 1–2 s per question	engineering result
Frames add substantially over transcript alone	v1 LVB +9.1 pts; MIT (debiased) +15.5 pts	robust (CIs exclude 0)
Small learned selectors do not beat transcript relevance on this lecture data	Head B, residual scorers, completion head, local-feature head	negative result, controlled
The last frame slot has measurable headroom that no tested selector captures	exhaustive fourth-frame table: oracle 45.2% vs 38.6%	measured on development data
Answerer-conditioned question filtering amplifies option-position bias	gold A/B 171/197; C/D-gold dropped 61% vs 39%	methodological finding
A new frame-selection method that beats baselines	—	not claimed

2. System design

2.1 Version 1 (branch QAframework)

Transcript lines are grouped into 20-second units and ranked with BM25 against the question and its options. The top windows (up to 4, expanded by one neighbour) define where to look and give a retained excerpt of at most 120 words. Frames are then chosen by one of five strategies (Section 5.1) and passed, with timestamp labels, to the frozen Qwen3-VL-2B-Instruct (revision 89644892, bfloat16, greedy decoding, 640-pixel frames). Answers cite timestamps, and every stage is timed.

2.2 Version 2 (branch framework-2)

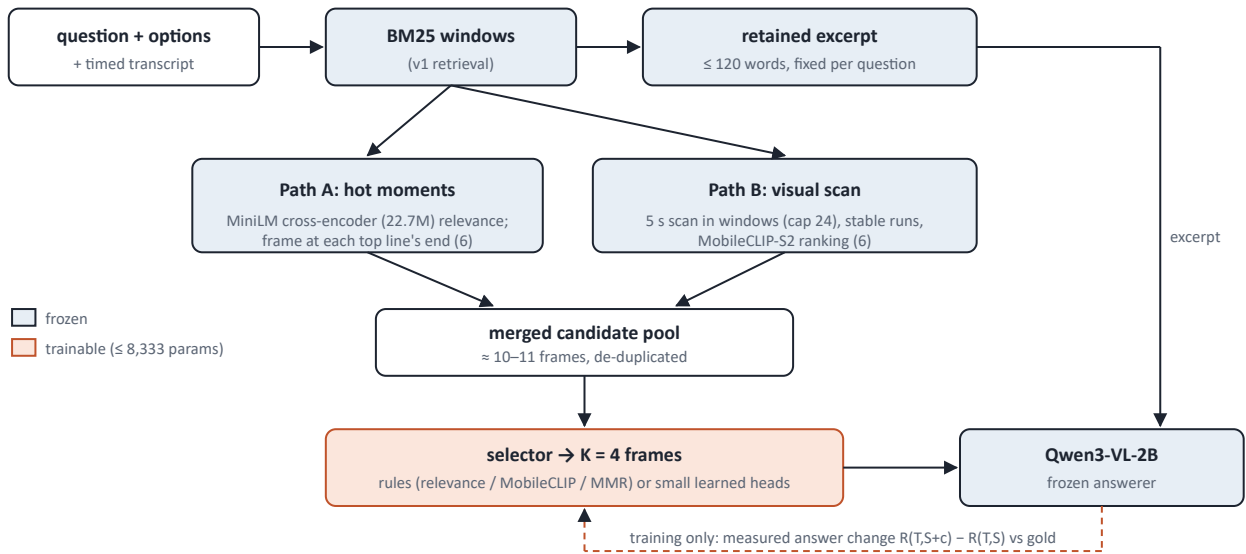


Figure 1. The v2 pipeline. All large models are frozen; only the selector can be trained, from labels produced by the same frozen answerer against gold answers (dashed).

v2 keeps the v1 retrieval and adds two proposal paths that feed one candidate pool per question:

- **Path A (transcript hot moments).** A frozen MiniLM-L6 cross-encoder (22.7M parameters) scores every transcript line inside the windows. The six best lines each contribute a frame taken 0.3 s before the line ends, when the board is fullest. In the main experiment these proposals use the pretrained relevance score; the trained Head A only re-ranks the fixed pool.
- **Path B (visual scan).** Frames are decoded every 5 s inside the windows (at most 24). Runs of visually similar frames (8×8 average hash within distance 6) collapse to their last frame, and MobileCLIP-S2 ranks the survivors against the question; the top six enter the pool. An opt-in *balanced* policy spreads the same 24 slots across all windows instead of filling the earliest first.
- **Merge.** Path B frames within 1 s or hash distance 3 of a Path A frame are merged into it. Pools hold about 10–11 frames. Each candidate gets one MobileCLIP embedding, RapidOCR text and nearby speech.
- **Selector.** Picks $K = 4$ frames. Rules: MobileCLIP top-4 (A), MMR diversity (B), zero-shot transcript relevance (F). Learned: Head B, residual scorers, completion heads (Section 4).

Component	Model	Size	Status
Answerer and label source	Qwen3-VL-2B-Instruct	2B	frozen
Frame / text embeddings	MobileCLIP-S2 (open_clip, datacompdr)	99.2M (35.8M image, 63.4M text)	frozen
Transcript relevance	cross-encoder/ms-marco-MiniLM-L6-v2	22.7M	frozen
On-screen text	RapidOCR (ONNX, CPU)	—	frozen
Head A (speech vs frame credit)	linear head on 6 features	14 params	trained
Head B (frame utility)	3 projections + MLP	8,333 params (15,257 pilot)	trained
Residual scorers	pairwise ridge around MobileCLIP	17 / 27 coefficients	trained
Completion heads	pairwise ridge around relevance	26 / 34 coefficients	trained

3. Datasets

3.1 LongVideoBench subset

440 validation questions on 252 videos (8 s to 60 min) with subtitles, 4–5 options, chance level 21.4%. Gold positions are naturally balanced (105/103/92/83/57). This subset was used to develop and benchmark v1; it is not the full benchmark (1,337 validation questions) and not an untouched test.

3.2 MIT lecture questions (constructed)

Existing benchmarks rarely require local speech *and* board content, so we built a lecture set:

1. **Lectures.** 20 MIT OpenCourseWare lectures by Gilbert Strang (11 from 18.06SC, 9 from 18.065), CC BY-NC-SA 4.0, with captions, selected in a fixed sha256 order.
2. **Mining.** Each lecture is cut into 30-second windows. A larger generator, Qwen3-VL-4B-Instruct (deliberately not the 2B answerer), sees two frames (window middle and end) and the captions, labels the speech/board relation and writes one multiple-choice question with four options. 591 windows were processed; 589 parsed.
3. **Verification.** Rule checks: quoted speech must occur in the captions, the answer must be grounded in the speech for speech-bearing relations, options must be distinct (25 rejected). Then the frozen 2B answerer is asked under four conditions: no evidence, captions only, frame only, both. 50 items failing both single-source conditions were rejected; 349 passed.
4. **Prior filter.** Questions the 2B answerer got right with no evidence (152) were dropped, leaving **197 questions** (79 speech-and-board, 74 speech, 44 board-support; 4–17 per lecture).

All items are marked synthetic and are not human-verified. Section 7 shows that step 4, combined with the generator's habit of putting the answer first, created a strong option-position bias.

4. Labels, learned selectors and protocol

Utility labels. $R(q, T, S) = 1$ if the frozen answerer answers question q correctly with retained transcript T and frame set S . Gold answers are used only to compute labels, never as inputs.

- **Head B:** $R(T, S + c) - R(T, S)$ for single frames on an empty history: 1,573 rows from 197 questions (1,249 zero, 237 positive, 87 negative), 9 answerer calls per question.
- **Head A:** separate interventions on an empty context, line alone vs frame alone, for 1,182 Path A moments (116 positive text, 367 positive frame labels).

- **Fourth-frame completion:** the relevance baseline fixes three frames; every other candidate c is labelled $R(T, \text{top-3} + c)$. Exhaustive (1,500 completions, 1,368 fresh calls), so any fourth-frame policy is scored exactly offline, including an oracle.

Learned selectors. Head B (projected MobileCLIP, OCR and nearby-speech embeddings plus scalar and history features; regression with within-question ranking). Residual scorers: a pairwise ridge correction to MobileCLIP with 17 coefficients, or 27 with retained-transcript lexical interactions. Completion heads: a pairwise ridge correction to the relevance rank, with 26 global features (candidate plus redundancy and novelty against the chosen frames and speech), optionally plus 8 local features (question relevance of 8 full-coverage tiles and chalk-ink novelty against the chosen frames).

Protocol. Leave-one-lecture-out cross-validation: heads evaluating a lecture never saw its labels; ridge strength is chosen by inner leave-one-lecture-out with a no-change fallback. Every arm sees the same pool, retained transcript, answerer, prompt and $K = 4$. Intervals are 95% lecture- or video-level bootstrap. All answerer calls go through an exact-request cache, so call counts are exact. Latency is a sum of measured stage medians, not an end-to-end timing.

5. Results

5.1 Version 1 on LongVideoBench

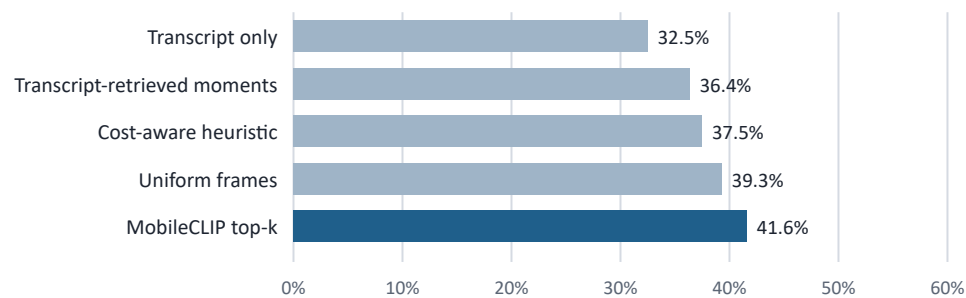


Figure 2. v1 accuracy on 440 LongVideoBench questions, at most 8 frames, same frozen 2B answerer. Chance is 21.4%.

Strategy	Accuracy	95% CI	Frames shown	Time / question
Transcript only	32.5%	28.3%–36.7%	0.0	1.14 s
Transcript-retrieved moments	36.4%	31.7%–40.8%	4.2	1.11 s
Cost-aware heuristic	37.5%	32.9%–42.0%	5.7	1.47 s
Uniform frames	39.3%	34.7%–44.0%	6.0	1.26 s
MobileCLIP top-k	41.6%	36.7%–46.4%	6.0	2.25 s

MobileCLIP top-k is the best v1 strategy (+9.1 points over transcript only, CI +4.6 to +13.6); its lead over uniform frames (+2.3) is not significant. Leaderboard 7–8B models score 40–50% with 8–16 frames on the full benchmark; this is a reference point, not a matched comparison.

5.2 Version 2 main experiment (MIT, original option order)

Arm (K = 4, 197 questions, 20 held-out lectures)	Accuracy	Composed latency
Transcript only	15.2%	—
A. MobileCLIP top-4	34.5%	4.48 s
B. MobileCLIP + MMR	36.0%	4.48 s
C. Head B with OCR (3 seeds)	34.7%	7.66 s
C. Head B without OCR	32.8%	4.48 s
C. Head B without OCR or speech features (corrected ablation)	32.5%	—
E. Trained Head A frame credit	30.5%	4.49 s
F. Zero-shot transcript relevance	38.6%	4.48 s
G. Trained Head A text credit	37.1%	4.49 s

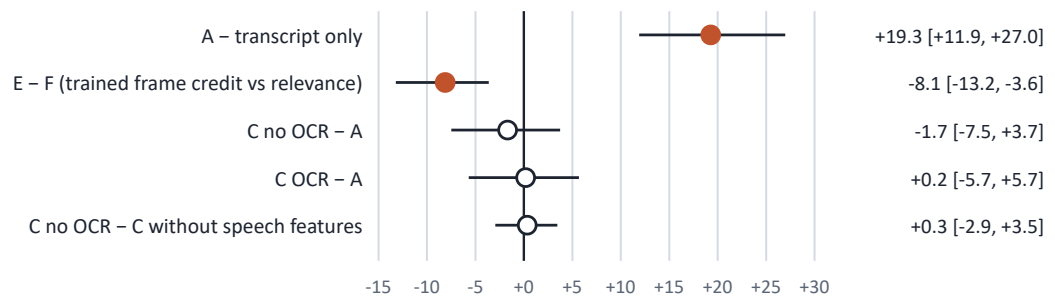


Figure 3. Paired differences (points) with 95% lecture-bootstrap intervals. Filled dots: interval excludes zero.

Frames matter; the learned Head B is not better than MobileCLIP and does not use its speech features; the trained frame-credit head is significantly worse than zero-shot relevance. Head A's out-of-fold AUC for frame gain was 0.50 (zero-shot 0.45). Stage medians: decode 2.11 s, OCR 3.15 s, answer 2.05 s.

5.3 Does speech change which frames help?

For 894 matched question/frame pairs measured with and without the retained transcript, positive frame-utility labels agree moderately (Cohen's $\kappa = 0.56$). Of 132 frames useful only without speech, 67 are redundant (speech already answers), 62 fail because the combined input is wrong although the frame alone was right, and 3 fail only in combination. These counts use the original option order and therefore include position-bias effects (Section 7).

5.4 Residual scorers and the fourth frame

Selector (K = 4)	Correct / 197	Accuracy
MobileCLIP	68	34.5%
MMR	71	36.0%
Zero-shot relevance	76	38.6%
Residual scorer, 17 coefficients	74	37.6%
Residual scorer + retained-transcript features	74	37.6%

Residual - MobileCLIP +3.0 (CI -1.0 to +7.3); residual - relevance -1.0 (CI -5.9 to +3.5).

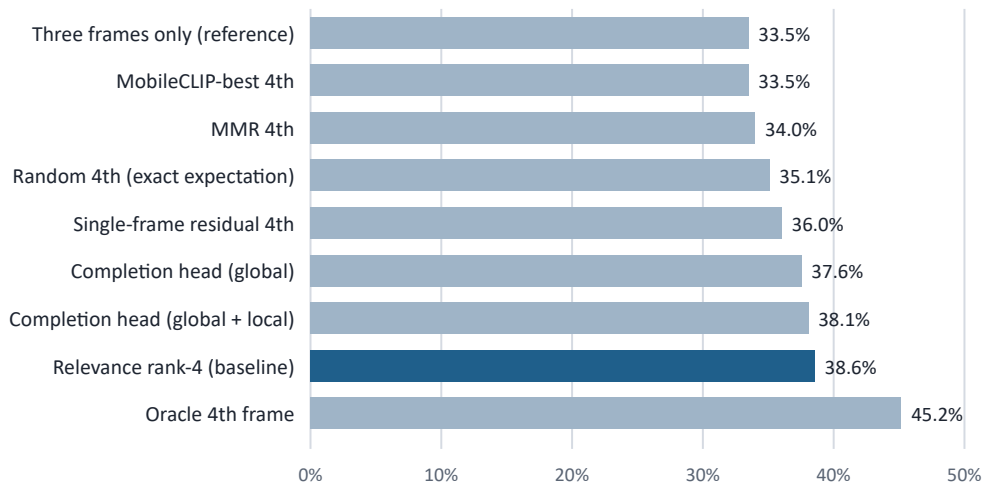


Figure 4. Exact four-frame accuracy of fourth-frame policies after the relevance baseline fixes three frames (exhaustive table, no extra calls per policy).

The deployment-matched completion head scores 37.6% (-1.0, CI -2.5 to +0.0) and falls back to the baseline in 18/20 folds; adding local board features gives 38.1% (-0.5, CI -1.6 to +0.0), fallback in 19/20. The oracle is +6.6 points above the baseline (CI +3.6 to +9.9), but the choice changes the answer in only 36 of 197 questions: too few to learn from. In 13 of 66 questions the three chosen frames answer correctly and some fourth frame breaks the answer; in 23 of 131 a fourth frame rescues it. MobileCLIP (-5.1) and MMR (-4.6) completions are significantly worse than relevance in this setting.

5.5 Candidate generation: balanced scan, crop and de-duplication

Change (only the pool or scout feature differs)	Data	Effect on MobileCLIP top-4
Balanced scan (original order)	MIT 197	+4.6 (CI -0.5 to +10.4)
Balanced scan (circular, debiased)	MIT 197	+3.6 (CI +0.4 to +6.9)
Balanced scan: declared transfer test	LongVideoBench 440	-0.5 (CI -3.2 to +2.3)
Whole frame padded to a square (same cost)	MIT balanced pools	-4.6 (CI -9.2 to -0.5)
Three full-coverage tiles (1.8× scout cost)	MIT balanced pools	+1.5 (CI -1.9 to +5.0)

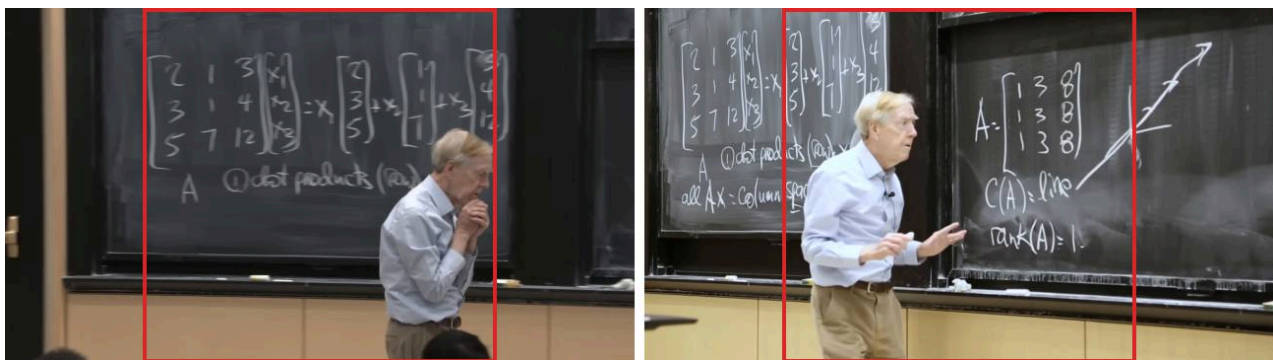


Figure 5. Two candidate frames from MIT 18.065 lecture 3. The red box is what MobileCLIP-S2's default transform keeps (resize to 256, centre crop): 56% of the width. The answerer always sees the full frame.

The balanced scan touches all 711 retrieved windows instead of 273 and helped on MIT, but it did not transfer to LongVideoBench, where relevance was also worse than MobileCLIP (-2.3, CI -5.5 to +0.9). Padding shrinks writing and hurts; tiles change 75% of selections without a reliable gain. Re-decoding every scan frame shows the stable-run rule drops 895 of 4,728 frames, almost all because the lecturer

moved; only 4 are small local edits, and the one flagged as lost writing is a chalk-tray reflection. Deduplication is not a measurable loss on these lectures.

6. Best method per setting

Setting	Best method	Score	Runner-up	Difference
LongVideoBench 440, v1, ≤ 8 frames	MobileCLIP top-k	41.6%	Uniform frames 39.3%	+2.3 (n.s.)
LongVideoBench 440, v2 pools, $K = 4$	MMR (legacy scan)	43.2%	MobileCLIP 43.0%	n.s.
MIT 197, original order, $K = 4$	Relevance (either scan) / MobileCLIP balanced	38.6% / 39.1%	Residual 37.6%	n.s.
MIT 197, circular (debiased), $K = 4$	MobileCLIP, balanced scan	53.3%	Relevance, balanced 52.4%	+0.9 (n.s.)
MIT fourth frame (fixed top-3)	Relevance rank-4	38.6%	Completion head + local 38.1%	-0.5

No learned selector is the best in any setting. The strongest choices are label-free rules, and which rule wins depends on the data: transcript relevance on chalkboard lectures, image similarity on general long videos.

7. Validity problems found and corrected

7.1 Option-position bias

The generator never shuffled options and its example used answer index 0, so gold is A or B in 282 of 349 verified questions. The prior filter used the same 2B answerer; it prefers C/D, so C/D-gold questions were "answered without evidence" more often and dropped more often (61% vs 39%). In the kept 197, gold counts by position are [79, 92, 24, 2] while the transcript-only model picks [38, 17, 60, 82]. "Always B" would score 46.7%. We re-evaluated the main arms under all four cyclic option rotations (circular evaluation, 3,495 extra calls):

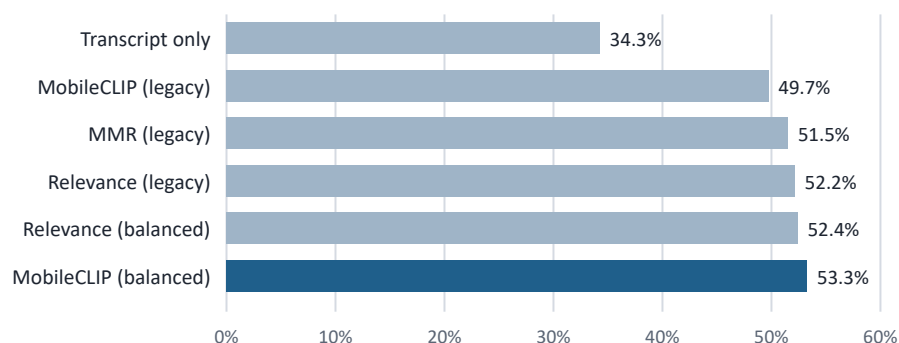


Figure 6. MIT accuracy after debiasing (mean over the four option rotations). Original-order values were about 15 points lower.

Debiasing raises transcript-only from 15.2% to 34.3% and every frame arm by about 15 points. Rankings keep their direction; relevance – MobileCLIP becomes +2.4 (n.s.); balanced – legacy for MobileCLIP becomes significant on MIT (+3.6, CI +0.4 to +6.9) but, as shown in 5.5, did not transfer. Label-based analyses (5.3, 5.4) used the original order and carry this caveat.

7.2 Other corrections

- **Interpretation:** an early claim that speech "mostly makes frames redundant" was wrong; about half the cases are failures of the combined input (5.3).
- **Speech ablation:** the first "no speech features" Head B still received speech-derived scores; corrected and rerun: 32.5% vs 32.8%.
- **Cache identity:** answer-cache keys omitted frame time labels, transcript ids/end times and decoding settings. The key now hashes the full rendered request and stores predictions, not correctness. An audit of 10,609 historical records found no possible collision; historical results replay exactly.
- **Data checks:** completion labels reject duplicates and inconsistent anchors; video leakage is checked on the whole manifest before split filtering; pool comparisons assert that only the scan differs.

8. Relation to prior work and novelty

We do not claim a new method. Learning frame selection from a frozen answer model's signal (SeViLA, Frame-Voyager, TSPO, FrameOracle, question-aware synthetic keyframe supervision 2603.14953), history-conditioned selection (ReFoCUS, MarKey, FORTE, GIFT), dual subtitle and visual streams (VSI, Q-Gate) and spatio-temporal rationales (TranSTR) are all established. Learning the marginal benefit of the next item given the current list goes back to Ross et al. (ICML 2013). Our MMR completion rule is not an implementation of MarKey or FORTE, and its failure here does not refute them.

What this project adds is narrower: (1) an exhaustive, deployment-matched fourth-frame table for one frozen answerer, which gives exact offline scores for any policy and an exact oracle; (2) controlled negative results for four families of small selectors trained on such labels; (3) a concrete, reproducible case in which answerer-conditioned filtering of synthetic questions amplifies option-position bias enough to invert conclusions about absolute accuracy. Option-position bias itself is well known (circular evaluation was introduced for it); the amplification mechanism and its size in a video-QA construction pipeline are the specific observation.

9. Limitations

- The lecture set is synthetic, not human-verified, from one course and one lecturer, and has been used for many development decisions; it is not a final test set.
- The same frozen 2B model produced labels and is evaluated; a second answerer has not been tested.
- Latency figures are composed stage medians, not end-to-end timings; some were measured while another GPU job ran.
- The LongVideoBench subset is not the full benchmark and was used to develop v1.
- Learned heads were trained on at most 1,573 labels; only 36 questions carry fourth-frame signal.
- No published learned selector was reproduced as a baseline.

10. Next steps

1. **An untouched, properly built test set:** regenerate lecture questions with seeded option shuffling, check gold-position balance before and after any filter, and audit a sample by hand.

2. **EduVidQA (EMNLP 2025)**: 3,909 training and 269 expert-verified real test questions on video-disjoint splits; needs video access (0 of 296 videos local, transcripts for 87) and a validated free-text scorer.
3. **More signal for the last slot**: the +6.6-point headroom is real, but it sits in 36 questions; more varied training questions are needed before any learned selector can be judged.

Appendix: reproducibility and compute

Code: `src/videoqa/v2/`; experiments: `scripts/v2_*.py`; every number above is read from `reports/` by `scripts/build_report.py`. Hypotheses for the completion, local-feature and LongVideoBench experiments were committed before their results (`docs/v2/novelty_check.md`, `docs/v2/lvb_transfer.md`). 205 unit tests pass (CPU only).

Experiment	Fresh answerer calls	GPU time
v2 main (pools, labels, evaluation)	—	3.4 h
Residual four-frame QA	225	7 min
Fourth-frame completion labels	1,368	43 min
Balanced scan pools and comparison (MIT)	579	≈ 35 min
Scout crop ablation	563	≈ 20 min
Circular evaluation	3,495	1.9 h
LongVideoBench transfer (pools + comparison)	2,375	≈ 1.6 h

Hardware: one NVIDIA RTX 5060 Ti (16 GB), Windows 11, Python 3.12, PyTorch 2.11 (CUDA 12.8), transformers 5.17.